

THE KENYA NATIONAL EXAMINATION COUNCIL

KCPE 2003

MATHEMATICS

Time:2 hours

READ THESE INSTRUCTION CAREFULLY

1. You are given this question booklet and a separate answer sheet. The question booklet contains 50 questions
2. Do any necessary rough work in this booklet
3. When you have chosen your answer, mark it on the **ANSWER SHEET**, not in this question booklet

HOW TO USE THE ANSWER SHEET

4. Use only ordinary pencil.
5. Mark sure that you have written on the answer sheet:

YOUR INDEX NUMBER

YOUR NAME

NAME OF YOUR SCHOOL

6. By drawing a dark line inside the correct numbered boxes mark your full index number
7. Do not make any mark outside the boxes.
8. Keep the sheet as clean as possible and do not fold it.
9. For each of the questions 1-50 four answers are given. The answers are lettered A,B,C and D
10. On the answer sheet the correct answer is to be shown by drawing a dark line inside the box in which the letter have chosen. Chose the correct answer.

Example

In the question booklet:

9. What is the next number in the pattern:

1.2,2.4,4.8, _____?

On the answer sheet:

The answer is D

9. [A] [B] [C] [D] 19 [A] [B] [C] [D] 29 [A] [B] [C] [D] 39 [A] [B] [C] [D]

In the set, the box with the letter D printed in it is marked.

11. Your dark line **MUST** be within the box.
12. For each question **ONLY ONE** box is to be marked in each of the four boxes.

The Kenya national Examination Council,2003

1. What is the place value of digit 3 in the number 780402?
 - A. Hundred thousands
 - B. Ten thousands
 - C. Thous
 - D. Hundreds
2. What is the value of $3(5^2 - 4^2) + 51 \div 17$?
 - A. $4 \frac{10}{17}$
 - B. 9
 - C. 30
 - D. 62
3. Florence bought 18 bags of fruits at sh. 300 per bag. She spent sh. 700 on transport. The fruits in two bags got spoilt and she sold the rest at sh. 520 per bag. How much profit did she make?
 - A. SH. 8320
 - B. Sh. 3260
 - C. Sh. 2920
 - D. Sh. 2220
4. Which one of the following is 30030030 in words?
 - A. Thirty million three hundred and thirty
 - B. Thirty million three thousand and thirty
 - C. Thirty million thirty thousand and thirty
 - D. Thirty million three hundred thousand and thirty
5. What is the value of $\frac{0.31 \times 1.02}{1.2}$?
 - A. 263.5
 - B. 2.635
 - C. 0.2635
 - D. 0.022635
6. Wasilwa bought the following items from a shop

6kg of sugar @ sh. 45.00

$\frac{1}{2}$ kg of tea for sh. 90.00

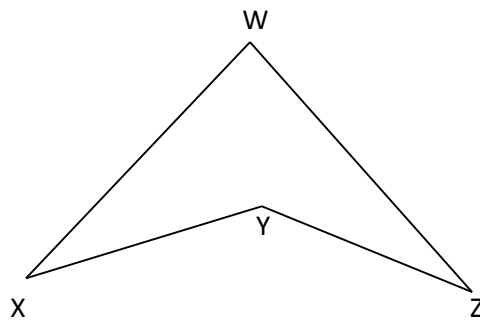
3kg of rice @ sh. 30.00

2kg of cooking fat @ sh. 70.00

If he used a one thousand shilling note to pay for items, what balance did he receive?

 - A. Sh. 410
 - B. Sh. 455
 - C. Sh. 590
 - D. Sh. 765
7. What is 468.3894 rounded off to the nearest tenth?
 - A. 470
 - B. 468.3

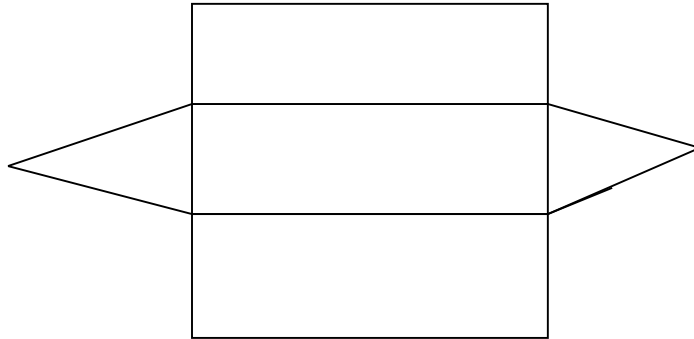
- C. 468.36
D. 468.4
8. Kamau wants to get change for sh. 1 000 note. The change is to consist of sh. 500, sh. 200, sh. 100 and sh. 50 notes. What is the smallest number of notes he can get?
A. 4
B. 5
C. 6
D. 7
9. Oloo slept at 2315h on Sunday and woke up six hours later. On what day and time did he wake up?
A. Sunday 5.15am
B. Monday 5.15p.m
C. Monday 5.15 a.m
D. Sunday 5.15 p.m
10. What is the difference in the value between the L.C.M of 8,12. And the G.C.D of 36 and 54?
A. 104
B. 84
C. 14
D. 6
11. What is the value of $\frac{7}{8}$ of $(\frac{4}{5} + \frac{1}{2}) \div \frac{1}{4}$
A. $9\frac{1}{320}$
B. $2\frac{7}{10}$
C. $6\frac{2}{5}$
D. $4\frac{4}{4}$
12. A rectangular container measures 75cm by 50cm by 1m. What is the capacity of the container in litres?
A. 3750L
B. 375L
C. 37.5L
D. 3.75L
13. What is the measure of the interior angle XYZ of the quadrilateral WXYZ constructed below?



- A. 220°
B. 140°

- C. 25°
- D. 20°

14. The figure below shows a net made up of three rectangles and two triangles



Which one of the following solids can be formed from this net?

- A. Rectangular prism
 - B. Triangular pyramid
 - C. Rectangular pyramid
 - D. Triangular prism
15. A plot of land is made up of a semi-circle and a rectangle. The semi-circle has a diameter of 14m while the rectangle measures 20m by 5m. What is the area of the plot?
(Take $\pi \frac{22}{7}$)
- A. 122m^3
 - B. 177m^3
 - C. 254m^3
 - D. 408m^3
16. A distance of 480m is represented on a scale drawing by line measuring 2.4cm. What is the scale used?
- A. 1:20
 - B. 1:200
 - C. 1:2000
 - D. 1:200000
17. Which one of the following represent the ratio 4:5 expressed as a percentage?
- A. $44 \frac{4}{9}\%$
 - B. $55 \frac{5}{9}\%$
 - C. 80%
 - D. 125%
18. The table below shows the number of times the team Simba, Moto and Dawa won, drew or lost in a competition. Three points were awarded for each game won, One point for each game drawn and no points for a game lost.

	Simba	Moto	Dawa
Won	2	4	3
Drawn	5	1	5
Lost	3	5	2

Which one of the following is order in which the team were ranked?

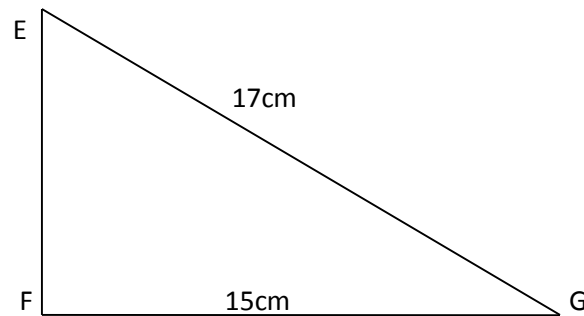
- | 1 st | 2 nd | 3 rd |
|-----------------|-----------------|-----------------|
| A. Simba | Moto | Dawa |
| B. Moto | Dawa | Simba |
| C. Dawa | Simba | Moto |
| D. Dawa | Moto | Simba |
19. A shopkeeper bought cooking fat in 30 one –kilogram packets and 20 five hundred gram packets She repacked the fats in packets of 125 grams each. How many packets did she get?
- A. 400
B. 320
C. 104
D. 32
20. The total number of pupils enrolled in schools in a districts was 48 000. In a pie chart the number enrolled in secondary schools was represented by an angle 120° while that in primary schools was represented by 150° . The rest were enrolled in the pre-schools. What was the number in the pre- schools?
- A. 36 000
B. 20 000
C. 16 000
D. 12 000
21. Which one of the following sets of measurements can be used to construct a right angled triangle?
- A. $4\frac{1}{2}$ cm , 6cm,9cm
B. $3\frac{3}{4}$ cm, $5\frac{1}{4}$ cm, 6cm
C. $2\frac{1}{4}$ cm, 3cm, $3\frac{3}{4}$ cm
D. $5\frac{1}{4}$ cm, 9cm, $11\frac{1}{4}$ cm
22. In one year the population of gazelles in a park was 3400. In the following year the population reduced by 25%. What was the size of the population of gazelles after the reduction?
- A. 850
B. 4250
C. 3375
D. 2550

23. The two parallel sides of a trapezium measure 16cm and 12cm. The perpendicular distance between the parallel lines is 8cm. What is the area of the trapezium?
- 112cm^2
 - 768cm^2
 - 224cm^2
 - 160cm^2
24. The length of two diagonals of a rhombus is 8cm and 6cm. What is the length of one side of rhombus?
- 3cm
 - 4cm
 - 5cm
 - 10cm
25. Amina paid sh. 24,000 for a TV set after getting a 20% discount. What was the marked price of TV set?
- Sh. 30 000
 - Sh. 28 000
 - Sh. 19 200
 - Sh. 20 000
26. In the figure below line EF is parallel to line GH. Line JK and line LM intersect at N angle $\text{NQH} = 120^\circ$ and angle $\text{NRF} = 160^\circ$

What is the size of angle PNS?

- 100°
 - 20°
 - 60°
 - 80°
27. Construct triangle PQR such that line $\text{qr} = 8\text{cm}$., Angle $\text{RPQ} = 55^\circ$ What is the length of line PQ?
- 6.8cm
 - 6.6cm
 - 13.7cm
 - 9.6cm
28. A saleslady earns a basic salary of sh. 4 000. For any sales above sh. 10 000, she is also paid a commission. In one month she sold items worth sh. 40 000 and earned a total of sh. 4900. What was her percentage commission?
- $2\frac{1}{4}\%$
 - 3%
 - $16\frac{1}{3}\%$
 - $22\frac{1}{2}\%$
29. Agola bought oranges and Mwangi bought $n + 1$ oranges Mueni bought twice as many oranges as both Agola and Mwangi. Which one of the following expressions shows the total number of oranges they bought?

- A. $6n + 3$
 B. $5n + 2$
 C. $4n + 2$
 D. $2n + 3$
30. Twenty members of group contributed sh. 80,000 each to buy a piece of land. They already had sh. 120 000 in a bank in which sh. 20 000 was to remain in the bank and the rest spent in buying the piece of land. How much did the piece of land cost?
- A. Sh. 1,720 000
 B. Sh. 1, 700 000
 C. Sh. 1, 600 000
 D. Sh. 180, 000
31. The diagram below represents triangle EFG in which line FG =15CM, Line EG =17cm and angle EFG = 90°



What is the area of the triangle?

- A. 60cm^2
 B. 68cm^2
 C. 120cm^2
 D. 127.5cm^2
32. Pupils in a school a shool prepared a circular flower garden of diameter 28 metres. They dug holes each 4 metres apart along the circumference for planting flowers. How many holes were dug?
 (Take $\pi \frac{22}{7}$)
- A. 154
 B. 88
 C. 11
 D. 22
33. Jane deposited sh. 2000 in a bank that paid compound interest at the rate of 12% p.a. How much money was in her account at the end of two years?
- A. Sh. 2508.80
 B. Sh. 2480
 C. Sh.2240
 D. Sh. 508.80

34. What is the value of x in the equation?

$$\frac{3}{5}(x-5) + \frac{2}{5}(3x+5) = 10$$

- A. $6\frac{1}{9}$
- B. $5\frac{5}{9}$
- C. 5
- D. $10\frac{1}{2}$

35. The number of children at a film show was 49 more than that of adults. The entry fee for each child was sh. 15 while that of adults was sh. 30. The total amount of money paid by the children was 2,055.

How much money was paid by children and adults altogether?

- A. Sh. 7635
- B. Sh. 4695
- C. Sh. 3375
- D. Sh. 2640

36. A cylindrical block has a diameter of 14cm and a height of 10cm by 5cm. What is the difference in their volume?

(Take $\pi \frac{22}{7}$)

- A. 310cm^3
- B. 790cm^3
- C. 2290cm^3
- D. 5410cm^3

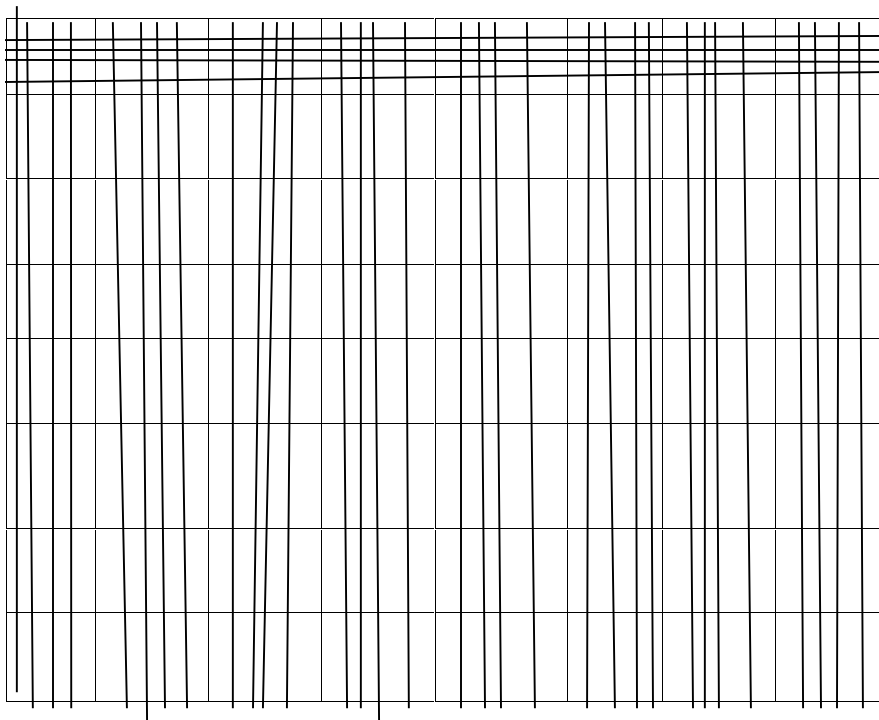
37. A clock gains 12.5 second every hour. If it was set correctly at 8.00am on Sunday, what time will it shows when the correct time is 8.00am on Thursday

- A. 8.25am
- B. 8.20am
- C. 8.10am
- D. 8.05am

38. Kaole, MWasi and Mbithe shared some money. Mwasi got $\frac{2}{7}$ of the money and Mbithe got $\frac{2}{3}$ of the remainder what fraction did kaole receive?

- A. $\frac{16}{21}$
- B. $\frac{11}{21}$
- C. $\frac{5}{21}$
- D. $\frac{1}{21}$

39. THE graph below shows a school advisor's Journey from school W to P through school L.K and M



Between which two schools was she travelling at the heighest speed?

- A. M and P
- B. K and M
- C. L and K
- D. W and L

40. An open cylinder has aradius of $3\frac{1}{2}$ cm and a height of 10cm. What is its surface area?

(Take π 22/7)

- A. 385cm^2
- B. 220cm^2
- C. $285\frac{1}{2}\text{cm}^2$
- D. 296cm^2

41. The table below shows the postal rates for sending a ,money order

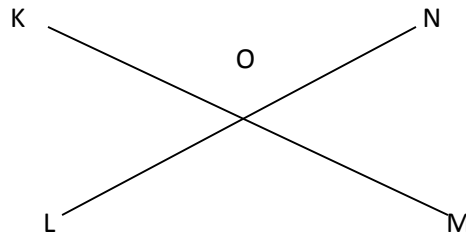
Mambo sent two money orders one of sh. 17, 500 and the other for 10,000

How much money did he pay at the post office altogether?

- A. Sh. 736
- B. Sh. 27,500
- C. Sh. 28,236
- D. Sh. 28 117

VALUE OF ORDER	COMMISSION
Not exceeding sh. 500	Sh. 42
501 -1000	Sh.113
1001 -3000	Sh.174
3001 -5000	Sh.209
5001 -10,000	Sh.295
10,001 -20,000	Sh. 441
20,001 -30,000	Sh.617

42. Below is part of construction of a quadrilateral KLMN with diagonals meeting at centre O. Complete the construction and draw a perpendicular line from L to meet KN at P. What is the length of LP?



- A. 7.8cm
B. 6.7cm
C. 4.6cm
D. 4.4cm
43. Muya earns sh. 5400 after working for 24 days. How much money should he receive if he is absent for 6 days?
A. Sh. 1350
B. Sh. 4050
C. Sh. 7200
D. Sh.6750
44. Three pupils Furaha, Gitahi and Komen contributed a total of sh. 400 for a party. Furaha contributed sh. 30 more than Gitahi while Komen contributed x, which one of the equations below can be used to find Komen's contribution
A. $5x - 90 = 1200$
B. $5x - 90 = 400$
C. $5x + 90 = 1200$
D. $7x + 30 = 400$
45. Katana's car uses one litre of petrol to cover 12.5km while simiyu's car uses 1 litre to cover 96km. If each covers 60km, how much more petrol does Simiyu's car use than Katana?
A. 11.05L
B. 6.25L
C. 4.80L
D. 1.45L
46. Given that $MK = \frac{33 \times 0.4}{8}$ and $K = 16.5$ What is the value of M?
A. 0.1

- B. 1
C. 0.01
D. 10
47. The floor of a square room is covered completely by four equal carpets. The area of each carpet is $3\frac{22}{48}\text{m}^2$
What is the length of one side of the room in metres?
A. $6\frac{44}{9}$
B. $7\frac{3}{7}$
C. $3\frac{5}{7}$
D. $1\frac{6}{7}$
48. The hire purchase terms of cupboard is a deposit of sh.4,400 and six monthly installments of sh. 900 each. The hire purchase price is 175% of the cost price while the cash price is 25% more than the cost price
What is the cash price of the cupboard?
A. Sh. 9,800
B. Sh. 5 600
C. Sh. 7 840
D. Sh. 7,000
- 49.

Which one of the following shapes can complete the pattern above?

50.

